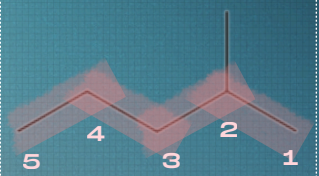
AO1 FOUNDATION C_6H_{14}

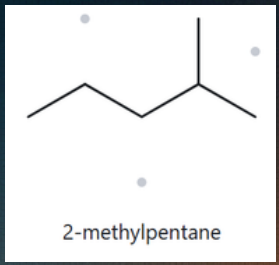
NUMBERED PARENT



NAMING TRAIL

PARENT: LONGEST CONTINUOUS CHAIN = 5 CARBONS (PENTANE). THE LONGEST PATH RUNS FROM THE LEFTMOST TERMINAL CARBON TO THE RIGHTMOST TERMINAL CARBON.
NUMBERING: LEFT → RIGHT GIVES THE METHYL SUBSTITUENT C4, WHILE RIGHT → LEFT GIVES C2. CHOOSE RIGHT → LEFT BECAUSE 2 < 4.
SUBSTITUENTS: METHYL ON C2.

ALKANE CHECK



RETURNED NAME:
2-methylpentane

MATCH ✓

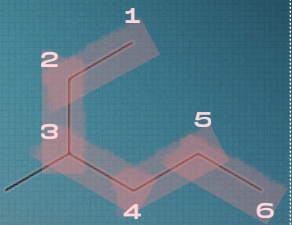
2-methylpentane

✓LONGEST CHAIN/RING CONFIRMED ✓LOWEST LOCANTS ✓ALPHABETICAL ORDER ✓PUNCTUATION ✓FORMULA COUNTED

A02 FOUNDATION

C_7H_{16}

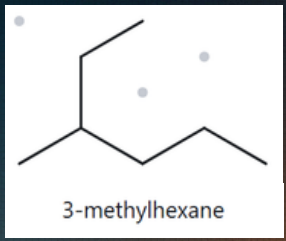
NUMBERED PARENT



NAMING TRAIL

PARENT: LONGEST CONTINUOUS CHAIN = 6 CARBONS (HEXANE). THE LONGEST PATH RUNS FROM THE UPPER TERMINAL CARBON DOWN TO THE RIGHTMOST TERMINAL CARBON.
NUMBERING: FROM THE UPPER END GIVES THE METHYL SUBSTITUENT C3; FROM THE RIGHT END GIVES C4. C3 IS CHOSEN.
SUBSTITUENTS: METHYL ON C3.

ALKANE CHECK



RETURNED NAME:
3-methylhexane

MATCH ✓

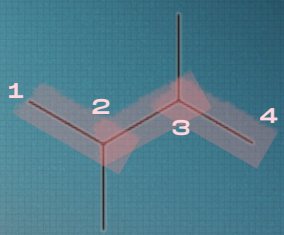
3-methylhexane

✓LONGEST CHAIN/RING CONFIRMED ✓LOWEST LOCANTS ✓ALPHABETICAL ORDER ✓PUNCTUATION ✓FORMULA COUNTED

AO3 FOUNDATION

C_6H_{14}

NUMBERED PARENT



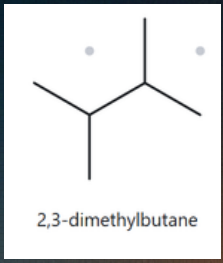
NAMING TRAIL

PARENT: LONGEST CONTINUOUS CHAIN = 4 CARBONS (BUTANE).

NUMBERING: EITHER DIRECTION GIVES THE SAME SUBSTITUENT POSITIONS, 2 AND 3.

SUBSTITUENTS: METHYL ON C2 AND C3.

ALKANE CHECK



RETURNED NAME:

2,3—dimethylbutane

MATCH ✓

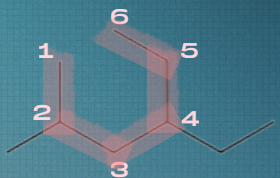
2,3—dimethylbutane

✓LONGEST CHAIN/RING CONFIRMED ✓LOWEST LOCANTS ✓ALPHABETICAL ORDER ✓ PUNCTUATION ✓FORMULA COUNTED

AO4 DEVELOPING

C_9H_{20}

NUMBERED PARENT



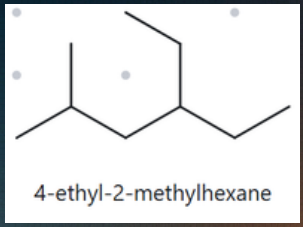
NAMING TRAIL

PARENT: LONGEST CONTINUOUS CHAIN = 6 CARBONS (HEXANE).

NUMBERING: LEFT → RIGHT GIVES SUBSTITUENT POSITIONS 2 AND 4. RIGHT → LEFT GIVES 3 AND 5. CHOOSE 2 AND 4.

SUBSTITUENTS: METHYL ON C2 AND EHTYL ON C2.

ALKANE CHECK



RETURNED NAME:

4-ethyl-2-methylhexane

MATCH ✓

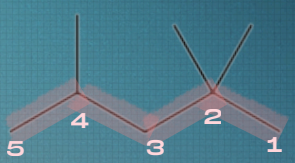
4-ethyl-2-methylhexane

✓LONGEST CHAIN/RING CONFIRMED ✓LOWEST LOCANTS ✓ALPHABETICAL ORDER ✓PUNCTUATION ✓FORMULA COUNTED

AO5 DEVELOPING

C_8H_{18}

NUMBERED PARENT



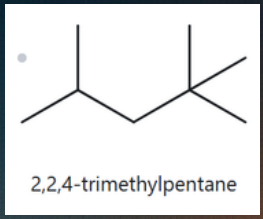
NAMING TRAIL

PARENT: LONGEST CONTINUOUS CHAIN = 5 CARBONS (PENTANE). THE LONGEST PATH RUNS FROM EITHER TERMINAL METHYL GROUP ON THE LEFT THROUGH THE CENTRAL CHAIN TO ONE OF THE TERMINAL METHYL GROUPS ON THE RIGHT.

NUMBERING: NUMBER FROM THE END NEAREST THE CARBON BEARING TWO METHYL SUBSTITUENTS. THIS GIVES LOCANTS 2,2,4 RATHER THAN 2,4,4

SUBSTITUENTS: METHYL ON C2 (TWO GROUPS) AND C4 (ONE GROUP)

ALKANE CHECK



RETURNED NAME:
2,2,4–trimethylpentane

MATCH ✓

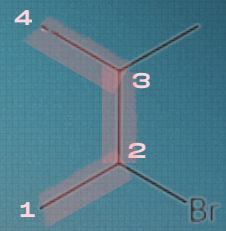
2,2,4–trimethylpentane

✓LONGEST CHAIN/RING CONFIRMED ✓LOWEST LOCANTS ✓ALPHABETICAL ORDER ✓PUNCTUATION ✓FORMULA COUNTED

A06 DEVELOPING

$C_5H_{11}Br$

NUMBERED PARENT



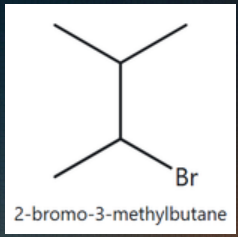
NAMING TRAIL

PARENT: LONGEST CONTINUOUS CHAIN = 4 CARBONS (BUTANE).

NUMBERING: BOTH DIRECTIONS GIVE LOCANTS 2 AND 3. IN THIS TIE, ALPHABETICAL ORDER DETERMINES THE NUMBERING. BROMO COMES BEFORE METHYL, SO BROMINE RECEIVES THE LOWER NUMBER, C2.

SUBSTITUENTS: BROMO ON C2 AND METHYL ON C3.

ALKANE CHECK



RETURNED NAME: 2,2,4–trimethylpentane

MATCH ✓

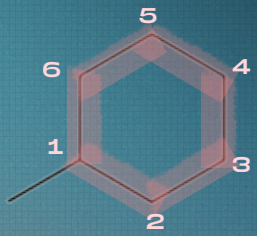
2–bromo–3–methylbutane

✓LONGEST CHAIN/RING CONFIRMED ✓LOWEST LOCANTS ✓ALPHABETICAL ORDER ✓PUNCTUATION ✓FORMULA COUNTED

A07 DEVELOPING

C_7H_{14}

NUMBERED PARENT



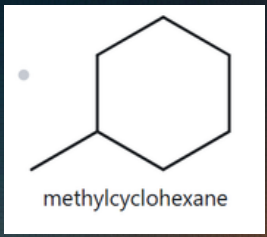
NAMING TRAIL

PARENT: SIX-MEMBERED CARBON RING = 6 CARBONS (CYCLOHEXANE). THE RING IS SELECTED AS THE PARENT BECAUSE IT CONTAINS MORE CARBONS THAN THE METHYL SUBSTITUENT.

NUMBERING: NO NUMBER IS NECESSARY BECAUSE THERE IS ONLY ONE SUBSTITUENT ON THE RING. ITS POSITION IS AUTOMATICALLY C1.

SUBSTITUENTS: METHYL ON C1.

ALKANE CHECK



RETURNED NAME:

methylcyclohexane

NOT MATCHING ✗

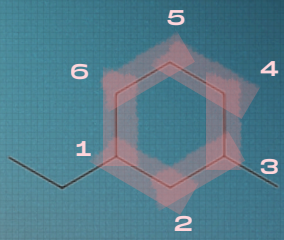
methylcyclohexane

✓LONGEST CHAIN/RING CONFIRMED ✓LOWEST LOCANTS ✓ALPHABETICAL ORDER ✓ PUNCTUATION ✓FORMULA COUNTED

A08 CHALLENGE

C_9H_{16}

NUMBERED PARENT



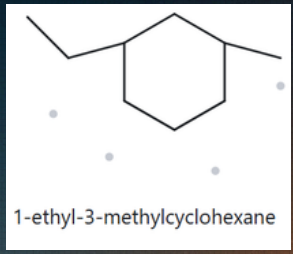
NAMING TRAIL

PARENT: SIX-MEMBERED CARBON RING = 6 CARBONS (CYCLOHEXANE).

NUMBERING: THE SUBSTITUENTS ARE SEPARATED BY ONE CARBON ON THE RING, GIVING LOCANTS 1 AND 3. BOTH NUMBERING DIRECTIONS PRODUCE 1,3. APPLY ALPHABETICAL ORDER: ETHYL COMES BEFORE METHYL, SO ETHYL RECEIVES C1.

SUBSTITUENTS: ETHYL ON C1 AND METHYL ON C3

ALKANE CHECK



RETURNED NAME:
1-ethyl-3-methylcyclohexane

MATCH ✓

1-ethyl-3-methylcyclohexane

✓LONGEST CHAIN/RING CONFIRMED ✓LOWEST LOCANTS ✓ALPHABETICAL ORDER ✓PUNCTUATION ✓FORMULA COUNTED

A09 CHALLENGE

$C_{10}H_{22}$

NUMBERED PARENT



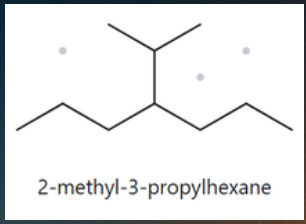
NAMING TRAIL

PARENT: LONGEST CONTINUOUS CHAIN = 7 CARBONS (HEPTANE). THE LONGEST PATH RUNS HORIZONTALLY FROM THE LEFTMOST TERMINAL CARBON TO THE RIGHTMOST TERMINAL CARBON.

NUMBERING: EITHER DIRECTION GIVES THE SUBSTITUENT POSITION C4.

SUBSTITUENTS: PROPAN-2-YL (ISOPROPYL) ON C4. THE BRANCH CONSISTS OF THREE CARBONS, WITH THE ATTACHMENT OCCURRING THROUGH ITS SECOND CARBON

ALKANE CHECK



RETURNED NAME:

2-methyl-3-propylhexane

NOT MATCHING ✗

4-(propan-2-yl)heptane

✓LONGEST CHAIN/RING CONFIRMED ✓LOWEST LOCANTS ✓ALPHABETICAL ORDER ✓PUNCTUATION ✓FORMULA COUNTED

A10 CHALLENGE

C_7H_{14}

NUMBERED PARENT



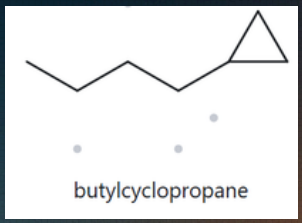
NAMING TRAIL

PARENT: LONGEST ACYCLIC CHAIN = 4 CARBONS (BUTANE). THE CYCLOPROPANE RING CONTAINS 3 CARBONS, SO THE OPEN CHAIN IS SELECTED AS THE PARENT.

NUMBERING: START FROM THE END NEAREST THE CYCLOPROPYL SUBSTITUENT, GIVING IT C1.

SUBSTITUENTS: CYCLOPROPYL ON C1.

ALKANE CHECK



RETURNED NAME:

butylcyclopropane

NOT MATCHING ✗

1-cyclopropylbutane

✓LONGEST CHAIN/RING CONFIRMED ✓LOWEST LOCANTS ✓ALPHABETICAL ORDER ✓PUNCTUATION ✓FORMULA COUNTED

B01 FOUNDATION

2-methylpropane

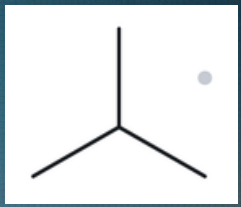
NAME DECODED

PARENT: PROPANE, A 3-CARBON CHAIN.

SUBSTITUENT: METHYL ON C2.

EXPECTED FORMULA: 3 + 1 = 4 C; ACYCLIC ALKANE → C_4H_{10}

BONE LINE STRUCTURE



FORMULA CHECK

COUNTED FROM THE DRAWING:
$CH_3-CH(CH_3)-CH_3$

- C = 4
- H = 3 + 1 + 3 + 3 = 10

RETURNED NAME:
2-methylpropane

MATCH ✓

✓ PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED

B02 FOUNDATION

3-ethylpentane

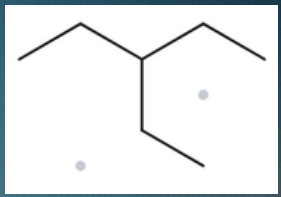
NAME DECODED

PARENT: PENTANE, A 5-CARBON CHAIN.

SUBSTITUENT: ETHYL (2 CARBONS) ON C3.

EXPECTED FORMULA: 5 + 2 = 7 C; ACYCLIC ALKANE → C_7H_{16}

BONE LINE STRUCTURE



FORMULA CHECK

COUNTED FROM THE DRAWING:
$CH_3CH_2-CH(CH_2CH_3)-CH_2CH_3$

- C = 5 + 2 = 7
- H = 16

RETURNED NAME:
3-ethylpentane

MATCH ✓

✓ PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED

B03 FOUNDATION

2,2–dimethylbutane

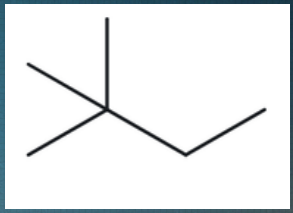
NAME DECODED

PARENT: BUTANE, A 4-CARBON CHAIN

SUBSTITUENT: TWO METHYL GROUPS ON C2

EXPECTED FORMULA: 4 + 2 = 6 C; ACYCLIC ALKANE → C_6H_{14}

BONE LINE STRUCTURE



FORMULA CHECK

COUNTED FROM THE DRAWING:
$CH_3–C(CH_3)_2–CH_2–CH_3$

- C = 4 + 2 = 6
- H = 3 + 6 + 2 + 3 = 14

RETURNED NAME:
2,2–dimethylbutane

MATCH ✓

✓ PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED

B04 DEVELOPING

3-ethyl-2-methylpentane

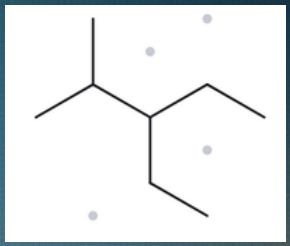
NAME DECODED

PARENT: PENTANE, A 5-CARBON CHAIN

SUBSTITUENT:
- METHYL ON C2.
- ETHYL ON C3.

EXPECTED FORMULA: 5 + 1 + 2 = 8 C; ACYCLIC ALKANE → C_8H_{18}

BONE LINE STRUCTURE



FORMULA CHECK

COUNTED FROM THE DRAWING: $CH_3-CH(CH_3)-CH(CH_2CH_3)-CH_2-CH_3$

- C = 5 + 1 + 2 = 8
- H = 18

RETURNED NAME:
3-ethyl-2-methylpentane
MATCH ✓

✓ PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED

B05 DEVELOPING

1-chloro-2-methylpropane

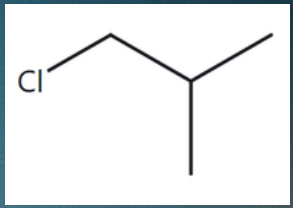
NAME DECODED

PARENT: PROPANE, A 3-CARBON CHAIN.

SUBSTITUENT:
- CHLORO ON C1.
- METHYL ON C2.

EXPECTED FORMULA: 3 + 1 = 4 C. REPLACING ONE HYDROGEN WITH CHLORINE GIVES C_4H_9CL.

BONE LINE STRUCTURE



FORMULA CHECK

COUNTED FROM THE DRAWING:
$CLCH_2-CH(CH_3)-CH_3$
- C = 3 + 1 = 4
- H = 2 + 1 + 3 + 3 = 9
- CL = 1

RETURNED NAME:
1-chloro-2-methylpropane

MATCH ✓

✓ PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED

B06 DEVELOPING

1,2–dimethylcyclopentane

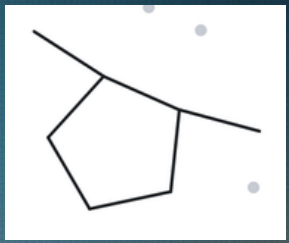
NAME DECODED

PARENT: CYCLOPENTANE, A 5-CARBON RING.

SUBSTITUENT: METHYL ON C1 AND C2.

EXPECTED FORMULA: 5 + 2 = 7 C; ONE RING → C_7H_{14}

BONE LINE STRUCTURE



FORMULA CHECK

COUNTED FROM THE DRAWING:

- CYCLOPENTANE RING: 5 C.
- TWO METHYL GROUPS: 2 C.
- TOTAL: 7 C.

RETURNED NAME:

1,2–dimethylcyclopentane

MATCH ✓

✓ *PARENT LENGTH MATCHES* ✓ *SUBSTITUENT POSITIONS MATCH* ✓ *FORMULA AGREES* ✓ *BACK-CHECK RECORDED*

B07 DEVELOPING

4–ethyl–3.3–dimethylheptane

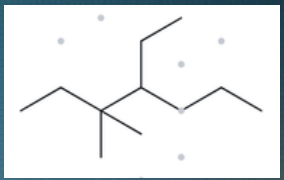
NAME DECODED

PARENT: HEPTANE, A 7-CARBON CHAIN

SUBSTITUENT:
- TWO METHYL GROUPS ON C3.
- ETHYL ON C4.

EXPECTED FORMULA: 7 + 2 + 2 = 11 C; ACYCLIC ALKANE → $C_{11}H_{24}$

BONE LINE STRUCTURE



FORMULA CHECK

COUNTED FROM THE DRAWING:
- HEPTANE CHAIN: 7 C.
- TWO METHYL GROUPS: 2 C.
- ETHYL GROUP: 2 C.
- TOTAL: 11 C.
- H = 24.

RETURNED NAME:
4–ethyl–3.3–dimethylheptane

MATCH ✓

✓ *PARENT LENGTH MATCHES* ✓ *SUBSTITUENT POSITIONS MATCH* ✓ *FORMULA AGREES* ✓ *BACK-CHECK RECORDED*

B08 CHALLENGE

tert-butylcyclohexane

NAME DECODED

PARENT: CYCLOHEXANE, A 6-CARBON RING.

SUBSTITUENT: TERT-BUTYL, A 4-CARBON GROUP ATTACHED THROUGH ITS CENTRAL CARBON.

EXPECTED FORMULA: 6 + 4 = 10 C; ONE RING → $C_{10}H_{20}$.

BONE LINE STRUCTURE

FORMULA CHECK

RETURNED NAME:

B09 CHALLENGE

3-ethyl-1,1-dimethylcyclohexane

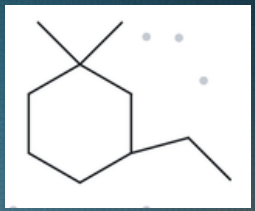
NAME DECODED

PARENT: CYCLOHEXANE, A 6-CARBON RING

SUBSTITUENT:
- TWO METHYL GROUPS ON C1.
- ETHYL ON C3.

EXPECTED FORMULA: 6 + 2 + 2 = 10 C; ONE RING → $C_{10}H_{20}$.

BONE LINE STRUCTURE



FORMULA CHECK

COUNTED FROM THE DRAWING:
- CYCLOHEXANE RING: 6 C.
- TWO METHYL GROUPS: 2 C.
- ETHYL GROUP: 2 C.
- TOTAL: 10 C.

RETURNED NAME:
3-ethyl-1,1-dimethylcyclohexane

MATCH ✓

✓ *PARENT LENGTH MATCHES* ✓ *SUBSTITUENT POSITIONS MATCH* ✓ *FORMULA AGREES* ✓ *BACK-CHECK RECORDED*

B10 CHALLENGE

2-ethylbutane

NAME DECODED

PARENT: BUTANE, A 4-CARBON CHAIN.

SUBSTITUENT: ETHYL (2 CARBONS) ON C2.

EXPECTED FORMULA: 4 + 2 = 6 C; ACYCLIC ALKANE → C_nH_{2n+2}

BONE LINE STRUCTURE

FORMULA CHECK

RETURNED NAME: